

Math Pretest

Due: Friday, September 18, end of day

Instructions

Set aside two hours for this pretest.

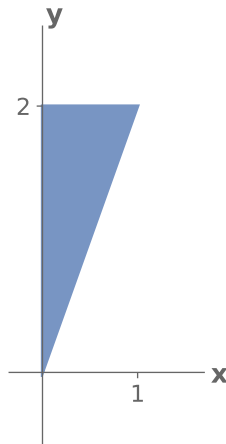
Please don't use a calculator, a computer, or any references. This pretest is meant to give both of us a clear picture of which areas you should practice more, so it won't help you to look things up. It is graded for completion, not correctness: doing it thoughtfully is what earns the credit.

Give your answers as expressions. For example, if your final answer to a question is $\sqrt{10}$, you should leave it as $\sqrt{10}$. If you have something that can easily be simplified, like e^0 or $\ln(1)$, you should simplify it.

1. Write the equation of a line with slope -5 that passes through the point $(-2, 2)$.
2. Find the slope and concavity of $f(x) = x^3 - x$ at $x = 2$.
3. Calculate the following derivatives:
 - a. $\frac{\partial f}{\partial y}$ for $f(x, y) = e^{3x \sin(ky)}$
 - b. $\frac{df}{dt}$ for $f(x, y) = x^3 e^y$, with $x(t) = \cos(2t)$ and $y(t) = t^2 + 1$
4. Evaluate the following integrals:

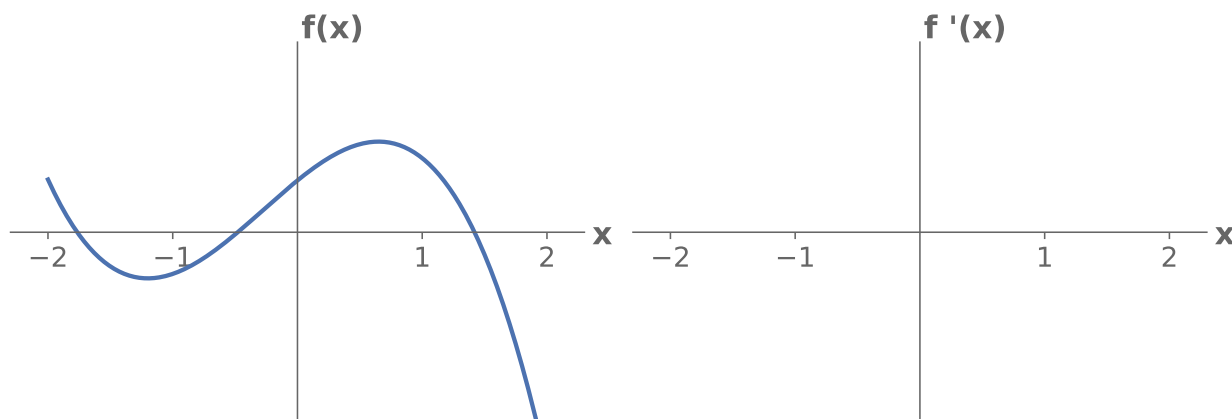
a. $\int \sin(x) [2 \cos(x) + 3]^5 dx$

- b. Integrate $f(x, y) = x - y$ over the triangular region shown below.

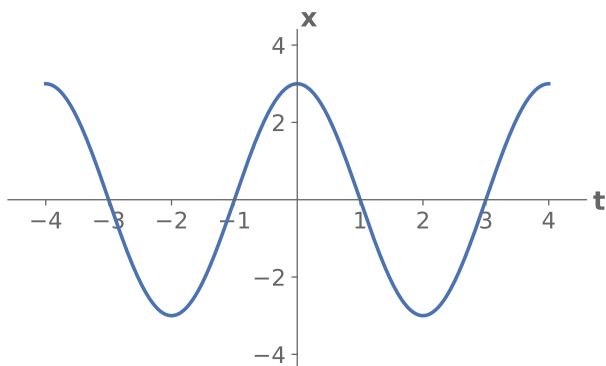


5. The point P in the x - y plane has Cartesian coordinates $x = -2$, $y = -2$. What are its polar coordinates ρ and ϕ ? (You may have learned these as r and θ — they mean the same thing.)

6. The plot below shows a function $f(x)$. On the axes provided at the right, sketch the function df/dx . Numbers are included on the horizontal axis, but you do not need to show numbers on the vertical axis.



7. The plot below shows a cosine function $x(t)$.
- What are the amplitude A , period T , and angular frequency ω ?
 - Write the function $x(t)$. *Hint: try plugging some points into your answer to make sure it matches the graph.*



8. The vector $\vec{A}$ is given by $\vec{A} = -2\hat{x} + 5\hat{y}$. The vector $\vec{B}$ has magnitude $2\sqrt{2}$ and direction 30° clockwise from the positive x axis.
- Find the magnitude and direction of $\vec{A}$.
 - Find the components of $\vec{B}$.
 - Find $\vec{A} \cdot \vec{B}$.
9. An object is moving in some complicated way, and you want to solve for its acceleration at some particular moment. You can't actually do that calculation here, because we're not giving you enough information. Instead, check each of the following possible answers to determine whether it has correct units. Assume that:
- F_1, F_2 , and so on are forces,
 - m is a mass,

- D is a distance,
- T is a time,
- v_0 and v_f are speeds.

Which of the following possible answers have correct units? (There may be more than one option with correct units.)

a. $a = \frac{v_0 + v_f}{T}$

b. $a = \frac{v_0 + v_f + 1}{T}$

c. $a = \frac{D}{T^2}$

d. $a = \frac{D}{T^2} e^{-v_f/v_0}$

e. $a = \frac{D}{T^2} e^{-v_f/(DT)}$